

Lab 1 Slide Deck

Number Systems and Base Conversion

Slide version of Lab 1.typ with worked answers, conversion tables, and math notation.

Coverage

- Counting in bases 8, 5, and 3
- Integer conversions across binary, octal, decimal, hexadecimal, base 6, and base 7
- Fractional conversions, significance of digits, and binary/hex grouping rules

Core Rules

- For integer conversion from decimal to base b , repeatedly divide by b and read remainders from bottom to top.
- For fractional conversion from decimal to base b , repeatedly multiply by b and read integer parts from top to bottom.
- For expansion to decimal, use weighted sums:

$$(d_n d_{n-1} \dots d_0 . d_{-1} d_{-2})_b = \sum_{k=0}^n d_k b^k + \sum_{k=1}^m d_{-k} b^{-k}.$$

Counting from 1 to $20_{\{10\}}$ in Other Bases

Decimal	Base 8	Base 5	Base 3
1	1_8	1_5	1_3
2	2_8	2_5	2_3
3	3_8	3_5	10_3
4	4_8	4_5	11_3
5	5_8	10_5	12_3
6	6_8	11_5	20_3
7	7_8	12_5	21_3
8	10_8	13_5	22_3
9	11_8	14_5	100_3
10	12_8	20_5	101_3

Counting from 11 to $20_{\{10\}}$ in Other Bases

Decimal	Base 8	Base 5	Base 3
11	13_8	21_5	102_3
12	14_8	22_5	110_3
13	15_8	23_5	111_3
14	16_8	24_5	112_3
15	17_8	30_5	120_3
16	20_8	31_5	121_3
17	21_8	32_5	122_3
18	22_8	33_5	200_3
19	23_8	34_5	201_3
20	24_8	40_5	202_3

Pattern

The place values in base b are $b^0, b^1, b^2, \dots$. Once the last single digit is reached, the next number increments the next place:
 $7_8 \rightarrow 10_8$, $4_5 \rightarrow 10_5$, and $2_3 \rightarrow 10_3$.

Method for Integer Conversion

To Decimal

Expand the numeral as a weighted polynomial in its base.

$$(a_m a_{m-1} \dots a_0)_b = a_m b^m + a_{m-1} b^{m-1} + \dots + a_0 b^0.$$

Example:

$$67_8 = 6 \times 8^1 + 7 \times 8^0 = 48 + 7 = 55_{10}.$$

From Decimal to Base b

Repeatedly divide by b :

- quotient goes to the next row,
- remainder becomes the next digit,
- final answer is read from the last remainder upward.

If a remainder is 10 to 15 in base 16, use

10 $\rightarrow$ A, 11 $\rightarrow$ B, 12 $\rightarrow$ C, 13 $\rightarrow$ D, 14 $\rightarrow$ E, 15 $\rightarrow$ F.

General Result

$N = (r_k r_{k-1} \dots r_1 r_0)_b$ where each r_i is the remainder sequence read in reverse order.

Convert 67_8 to Bases 6 and 7

Step 1: Convert to Decimal

$$67_8 = 6 \times 8^1 + 7 \times 8^0 = 48 + 7 = 55_{10}.$$

To Base 6

Dividend	Op	Base	Quotient	Remainder	Digit
55	÷	6	9	1	1
9	÷	6	1	3	3
1	÷	6	0	1	1

Answer

$$55_{10} = 131_6$$

To Base 7

Dividend	Op	Base	Quotient	Remainder	Digit
55	÷	7	7	6	1
7	÷	7	1	0	0
1	÷	7	0	1	6

Answer

$$55_{10} = 106_7$$

Convert $91_{\{10\}}$ to Bases 6 and 7

To Base 6

Dividend	Op	Base	Quotient	Remainder	Digit
91	÷	6	15	1	2
15	÷	6	2	3	3
2	÷	6	0	2	1

Answer

$$91_{10} = 231_6$$

To Base 7

Dividend	Op	Base	Quotient	Remainder	Digit
91	÷	7	13	0	1
13	÷	7	1	6	6
1	÷	7	0	1	0

Answer

$$91_{10} = 160_7$$

Convert $75_{\{16\}}$ to Bases 6 and 7

Step 1: Convert to Decimal

$$75_{16} = 7 \times 16^1 + 5 \times 16^0 = 112 + 5 = 117_{10}.$$

To Base 6

Dividend	Op	Base	Quotient	Remainder	Digit
117	÷	6	19	3	3
19	÷	6	3	1	1
3	÷	6	0	3	3

Answer

$$117_{10} = 313_6$$

To Base 7

Dividend	Op	Base	Quotient	Remainder	Digit
117	÷	7	16	5	2
16	÷	7	2	2	2
2	÷	7	0	2	5

Answer

$$117_{10} = 225_7$$

Convert 11001₂ to Bases 6 and 7

Step 1: Convert to Decimal

$11001_2 = 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 16 + 8 + 1 = 25_{10}.$

To Base 6

Dividend	Op	Base	Quotient	Remainder	Digit
25	÷	6	4	1	4
4	÷	6	0	4	1

Answer

$25_{10} = 41_6$

To Base 7

Dividend	Op	Base	Quotient	Remainder	Digit
25	÷	7	3	4	3
3	÷	7	0	3	4

Answer

$25_{10} = 34_7$

Decimal to Hexadecimal and Binary: $24_{\{10\}}$ and $454_{\{10\}}$

24_{10}

$$24 \div 16 = 1, \text{ remainder } 8$$

$$1 \div 16 = 0, \text{ remainder } 1$$

Therefore

$$24_{10} = 18_{16}.$$

Also,

$$24_{10} = 11000_2.$$

454_{10}

$$454 = 1 \times 16^2 + 12 \times 16 + 6$$

so

$$454_{10} = 1C6_{16}.$$

Binary by repeated division or from hex:

$$1 \rightarrow 0001, C \rightarrow 1100, 6 \rightarrow 0110.$$

Hence

$$454_{10} = 111000110_2.$$

Decimal	Hexadecimal	Binary
24	18_{16}	11000_2
454	$1C6_{16}$	111000110_2

Decimal to Hexadecimal and Binary: 6550_{10}

Repeated Division by 16

Division	Quotient	Remainder
$6550 \div 16$	409	6
$409 \div 16$	25	9
$25 \div 16$	1	9
$1 \div 16$	0	1

Result

Reading the remainders from bottom to top gives

$$6550_{10} = 1996_{16}.$$

Convert each hex digit to binary:

$$1 \rightarrow 0001, 9 \rightarrow 1001, 9 \rightarrow 1001, 6 \rightarrow 0110.$$

Therefore

$$6550_{10} = 1100110010110_2.$$

From Octal to Hex and Binary: 27_8 and 6325_8

27_8

$$27_8 = 2 \times 8^1 + 7 \times 8^0 = 16 + 7 = 23_{10}.$$

$$23_{10} = 17_{16} = 10111_2.$$

6325_8

$$\begin{aligned} 6325_8 &= 6 \times 8^3 + 3 \times 8^2 + 2 \times 8^1 + 5 \times 8^0 \\ &= 3072 + 192 + 16 + 5 = 3285_{10}. \end{aligned}$$

Repeated division by 16 gives

$$3285_{10} = \text{CD}5_{16}.$$

Hex to binary:

$$\text{C} = 1100, \text{D} = 1101, 5 = 0101,$$

so

$$6325_8 = 110011010101_2.$$

From Octal to Hex and Binary: Summary Table

Original	Decimal	Hex	Binary
27 ₈	23	17 ₁₆	10111 ₂
6325 ₈	3285	CD5 ₁₆	110011010101 ₂

Binary to Decimal and Hexadecimal

Binary	Decimal Expansion	Decimal	Hexadecimal
1101_2	$1 \times 2^3 + 1 \times 2^2 + 0 \times 2 + 1$	13	D_{16}
1000110_2	$1 \times 2^6 + 1 \times 2^2 + 1 \times 2^1$	70	46_{16}
111111_2	$2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$	63	$3F_{16}$

Why Hex is Fast

Group binary digits into chunks of four:

$$1101_2 = D_{16}, 1000110_2 = 01000110_2 = 46_{16}.$$

Each hexadecimal digit represents exactly 4 binary bits.

Most Significant and Least Significant Digits

Number	MSD	LSD	Reason
441.34	4	4	Leftmost non-zero digit is 4; rightmost written digit is 4.
0.245	0	5	Using the written form, the leftmost digit is 0 and the rightmost digit is 5.
234.00	2	0	The first digit is 2; the last written digit is the final 0 in the fractional part.

Interpretation

The **most significant digit** contributes the largest positional weight. The **least significant digit** contributes the smallest positional weight among the written digits. For fractional numerals, the LSD is often in the fractional tail.

Convert 204.125_{10} to Hexadecimal

Integer Part: 204

Division	Quotient	Remainder
$204 \div 16$	12	$12 = \text{C}$
$12 \div 16$	0	$12 = \text{C}$

Assemble the Answer

Integer part:

$$204_{10} = \text{CC}_{16}.$$

Fractional part:

$$0.125_{10} = .2_{16}.$$

Therefore

$$204.125_{10} = \text{CC}.2_{16}.$$

Fractional Part: 0.125

Multiplication	Product	Integer Part
0.125×16	2.0	2

Convert $631.25_{\{10\}}$ to Hexadecimal

Integer Part: 631

Division	Quotient	Remainder
$631 \div 16$	39	7
$39 \div 16$	2	7
$2 \div 16$	0	2

Assemble the Answer

Integer part:

$$631_{10} = 277_{16}.$$

Fractional part:

$$0.25_{10} = .4_{16}.$$

Hence

$$631.25_{10} = 277.4_{16}.$$

Fractional Part: 0.25

Multiplication	Product	Integer Part
0.25×16	4.0	4

Binary Fractions to Hexadecimal

Binary	Grouping	Hex	Reason
1001.1111 ₂	1001.1111	9.F ₁₆	Each 4-bit group maps directly to one hex digit.
110101.011001 ₂	00110101.01100100	35.64 ₁₆	Pad on the left of the integer part and on the right of the fractional part.
10100111011011 ₂	0010100111011011	29DB ₁₆	The grouped nibbles are 0010, 1001, 1101, 1011.

Grouping Rule

For the integer part, group bits from the right. For the fractional part, group bits from the left. Padding zeros do not change the value when they are added on the outer side of the grouping.

Hexadecimal to Binary

Hexadecimal	Nibble Expansion	Binary	Simplified Form
E ₁₆	E → 1110	1110 ₂	1110 ₂
1C ₁₆	1 → 0001, C → 1100	00011100 ₂	11100 ₂
A64 ₁₆	A → 1010, 6 → 0110, 4 → 0100	101001100100 ₂	101001100100 ₂
1F.C ₁₆	1 → 0001, F → 1111, C → 1100	00011111.1100 ₂	11111.11 ₂
239.4 ₁₆	2 → 0010, 3 → 0011, 9 → 1001, 4 → 0100	001000111001.0100 ₂	1000111001.01 ₂

IEEE 754 Single Precision Overview

Bit Layout

Field	Bits	Meaning	Width
Sign	s	Positive or negative	1
Exponent	e	Biased exponent	8
Fraction	f	Mantissa after the hidden leading 1	23

Stored Pattern

s **eeeeeeee** **ffffffffffffffffffffffff**

Value Formula

For a normalized 32-bit float,

$$\text{value} = (-1)^s \times (1.f)_2 \times 2^{e-127}.$$

Here 127 is the bias, so the true exponent is

$$E = e - 127.$$

Flow:

- Convert the decimal value to binary.
- Normalize it to the form $1.\text{xxxxxx} \times 2^E$.
- Store the sign bit.
- Add the bias to get the exponent field.
- Store the fractional part after the leading 1 in the mantissa.

Convert 13.25_{10} to IEEE 754 32-bit

Step 1: Convert to Binary

Integer part:

$$13_{10} = 1101_2.$$

Fractional part:

Multiply by 2	Result	Next Bit
0.25×2	0.5	0
0.5×2	1.0	1

Thus

$$0.25_{10} = 0.01_2$$

and

$$13.25_{10} = 1101.01_2.$$

Step 2: Normalize and Encode

Normalize:

$$1101.01_2 = 1.10101_2 \times 2^3.$$

So:

- sign bit $s = 0$
- actual exponent $E = 3$
- biased exponent $e = 3 + 127 = 130 = 10000010_2$
- mantissa $f = 10101000000000000000000$

Sign	Exponent	Fraction
0	10000010	10101000000000000000000

Final 32-bit pattern:

01000001010101000000000000000000

Convert -5.75_{10} to IEEE 754 32-bit

Step 1: Binary Form

Magnitude:

$$5_{10} = 101_2.$$

Fraction:

Multiply by 2	Result	Next Bit
0.75×2	1.5	1
0.5×2	1.0	1

Hence

$$0.75_{10} = 0.11_2$$

and

$$5.75_{10} = 101.11_2.$$

Step 2: Normalize and Encode

Normalize:

$$101.11_2 = 1.0111_2 \times 2^2.$$

So:

- sign bit $s = 1$ because the number is negative
- actual exponent $E = 2$
- biased exponent $e = 2 + 127 = 129 = 10000001_2$
- mantissa $f = 01110000000000000000000$

Sign	Exponent	Fraction
1	10000001	01110000000000000000000

Final 32-bit pattern:

11000000101110000000000000000000

Decode an IEEE 754 32-bit Pattern: Parse Fields

Example Pattern

Decode

01000001010101000000000000000000.

Field	Bits	Value	Interpretation
Sign	0	0	Positive number
Exponent	10000010	130	$E = 130 - 127 = 3$
Fraction	101010000000000000000000	0.10101_2	Hidden leading 1 gives 1.10101_2

Decode an IEEE 754 32-bit Pattern: Reconstruct Value

Reconstruct the Number

From the parsed fields:

- $\text{sign} = 0$
- $\text{exponent} = 10000010_2 = 130$, so $E = 3$
- $\text{significand} = 1.10101_2$

Apply the formula:

$$\text{value} = (-1)^0 \times 1.10101_2 \times 2^3.$$

Shift the binary point right by 3 places:

$$1.10101_2 \times 2^3 = 1101.01_2.$$

Convert to decimal:

$$1101_2 = 13, q \quad 0.01_2 = 0.25.$$

Therefore

$$01000001010101000000000000000000 = 13.25_{10}.$$